Probability Fundamentals

MDragon Data Tools - Study Guide

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1. Basic Probability Concepts

What is Probability?

Probability measures the likelihood of an event occurring, ranging from 0 (impossible) to 1 (certain).

Formula:

```
P(Event) = Number of favorable outcomes / Total number of possible outcomes
```

Key Terms

- **Experiment**: A process that produces outcomes
- **Sample Space (S)**: All possible outcomes
- **Event (E)**: A subset of the sample space

- **Complement (E')**: All outcomes NOT in E

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2. Probability Rules

Addition Rule

For mutually exclusive events A and B:

```
P(A or B) = P(A) + P(B)
```

For non-mutually exclusive events:

```
P(A or B) = P(A) + P(B) - P(A and B)
```

Multiplication Rule

For independent events A and B:

```
P(A and B) = P(A) × P(B)
```

For dependent events:

```
P(A and B) = P(A) × P(B|A)
```

Complement Rule

```
P(E') = 1 - P(E)
```

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3. Counting Techniques

Permutations (Order Matters)

```
nPr = n! / (n - r)!
```

where n = total items, r = items selected

Combinations (Order Doesn't Matter)

```
nCr = n! / (r! × (n - r)!)
```

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4. Step-by-Step Examples

Example 1: Simple Probability

📝 **Problem:** What is the probability of rolling a 4 on a fair six-sided die?

💡 **Solution:**

1. Identify favorable outcomes: {4} → 1 outcome
2. Identify total possible outcomes: {1, 2, 3, 4, 5, 6} → 6 outcomes
3. Apply formula:

   ```
   P(rolling 4) = 1/6 ≈ 0.167 or 16.7%
   ```

✅ **Answer:** 0.167 or 16.7%

Example 2: Addition Rule (Mutually Exclusive)

📝 **Problem:** A card is drawn from a standard deck. What is the probability of drawing a King OR a Queen?

💡 **Solution:**

1. P(King) = 4/52 = 1/13
2. P(Queen) = 4/52 = 1/13
3. These events are mutually exclusive (can't draw both at once)
4. Apply addition rule:

```
P(King or Queen) = P(King) + P(Queen)
                  = 1/13 + 1/13
                  = 2/13 ≈ 0.154
```

✅ **Answer:** 2/13 or approximately 15.4%

Example 3: Addition Rule (Non-Mutually Exclusive)

📝 **Problem:** A card is drawn from a standard deck. What is the probability of drawing a Heart OR a King?

💡 **Solution:**

1. P(Heart) = 13/52 = 1/4
2. P(King) = 4/52 = 1/13
3. P(Heart AND King) = 1/52 (King of Hearts)
4. These events are NOT mutually exclusive
5. Apply general addition rule:

```
P(Heart or King) = P(Heart) + P(King) - P(Heart and King)
                 = 13/52 + 4/52 - 1/52
                 = 16/52 = 4/13 ≈ 0.308
```

✅ **Answer:** 4/13 or approximately 30.8%

Example 4: Multiplication Rule (Independent Events)

📝 **Problem:** You flip a fair coin twice. What is the probability of getting heads both times?

💡 **Solution:**

1. P(Heads on first flip) = 1/2
2. P(Heads on second flip) = 1/2
3. Flips are independent
4. Apply multiplication rule:

```
P(Heads AND Heads) = P(Heads) × P(Heads)
                    = 1/2 × 1/2
                    = 1/4 = 0.25
```

✅ **Answer:** 0.25 or 25%

Example 5: Complement Rule

📝 **Problem:** The probability of rain tomorrow is 0.35. What is the probability of NO rain?

💡 **Solution:**

1. P(Rain) = 0.35
2. Apply complement rule:

```
P(No Rain) = 1 - P(Rain)
           = 1 - 0.35
           = 0.65
```

✅ **Answer:** 0.65 or 65%

Example 6: Combinations

📝 **Problem:** A committee of 3 people must be selected from a group of 8. How many different committees are possible?

💡 **Solution:**

1. n = 8 (total people)
2. r = 3 (people selected)
3. Order doesn't matter (combination)
4. Apply combination formula:

```
8C3 = 8! / (3! × 5!)
    = (8 × 7 × 6) / (3 × 2 × 1)
    = 336 / 6
    = 56
```

✅ **Answer:** 56 different committees

Example 7: Permutations

📝 **Problem:** How many different 3-digit lock codes can be created using digits 1-9 without repetition?

💡 **Solution:**

1. n = 9 (available digits)
2. r = 3 (digits in code)
3. Order matters (permutation)
4. Apply permutation formula:

```
9P3 = 9! / (9 - 3)!
    = 9! / 6!
    = 9 × 8 × 7
    = 504
```

✅ **Answer:** 504 different codes

5. Practice Problems

Problem 1

A bag contains 5 red marbles, 3 blue marbles, and 2 green marbles. What is the probability of randomly selecting a blue marble?

Problem 2

What is the probability of rolling a sum of 7 with two fair dice?

Problem 3

A student takes two exams. The probability of passing the first is 0.8 and passing the second is 0.7. If the exams are independent, what is the probability of passing both?

Problem 4

In a class of 30 students, 18 are female. A committee of 4 students is randomly selected. How many different committees are possible?

Problem 5

A password must contain 4 letters followed by 2 digits. Letters can be any of 26 letters (case-insensitive) and digits 0-9, with repetition allowed. How many different passwords are possible?

Answers to Practice Problems

1. **3/10 or 0.3** (3 blue out of 10 total marbles)
2. **1/6 or 0.167** (6 ways to roll 7 out of 36 total combinations)
3. **0.56** ($0.8 \times 0.7 = 0.56$)
4. **27,405** ($30C4 = 30!/(4!\times26!)$)
5. **45,697,600** ($26^4 \times 10^2$)

Quick Reference Formulas

Concept	Formula
Basic Probability	P(E) = favorable/total
Complement	P(E') = 1 - P(E)
Addition (Mutually Exclusive)	P(A or B) = P(A) + P(B)
Addition (General)	P(A or B) = P(A) + P(B) - P(A and B)
Multiplication (Independent)	P(A and B) = P(A) × P(B)
Permutation	nPr = n!/(n-r)!
Combination	nCr = n!/(r!×(n-r)!)

For more practice, use our Probability Calculator at MDragon Data Tools